

PART A — QUESTIONS

AF4-001

Question:

Solve: $6x + 5 = 41$

- A) 5
- B) 6
- C) 7
- D) 8

AF4-002

Question:

If $4n - 9 = 19$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF4-003

Question:

Simplify: $5(3x - 2) - 4x$

- A) $11x - 10$
- B) $15x - 10$
- C) $11x - 2$
- D) $19x - 10$

AF4-004

Question:

A shop charges \$14 per item plus a \$10 service fee. Which expression represents the total cost for k items?

- A) $14 + 10k$
- B) $14k + 10$
- C) $24k$
- D) $10k - 14$

AF4-005

Question:

If $f(x) = 6x - 5$, what is $f(3)$?

- A) 11
- B) 13
- C) 15
- D) 18

AF4-006

Question:

Which ordered pair satisfies the equation $y = 5x + 2$?

- A) (1, 6)
- B) (2, 12)
- C) (3, 17)
- D) (4, 24)

AF4-007

Question:

Solve: $5(x - 3) = 30$

A) 6

B) 7

C) 8

D) 9

AF4-008

Question:

A pattern begins: 2, 7, 12, 17, ... What is the next number?

A) 20

B) 21

C) 22

D) 23

AF4-009

Question:

Solve: $x/8 = 7$

A) 15

B) 42

C) 48

D) 56

AF4-010

Question:

A line has a slope of -3 and crosses the y-axis at 4. Which equation represents the line?

A) $y = 4x - 3$

B) $y = -3x + 4$

C) $y = 3x + 4$

D) $y = -4x + 3$

AF4-011

Question:

If 7 containers hold 84 pieces, how many pieces are in 11 containers at the same rate?

A) 112

B) 121

C) 132

D) 144

AF4-012

Question:

Solve: $20 - 4x = 8$

A) 2

B) 3

C) 4

D) 7

AF4-013

Question:

Which expression is equivalent to $14y - 6y + 3$?

- A) $8y + 3$
- B) $20y + 3$
- C) $8y - 3$
- D) $14y - 3$

AF4-014

Question:

A worker has already completed 18 tasks and completes 4 more tasks each hour. Which equation gives the total number of tasks T after h hours?

- A) $T = 18h + 4$
- B) $T = 4h + 18$
- C) $T = 22h$
- D) $T = 18 - 4h$

AF4-015

Question:

If $y = -3x + 12$, what is y when $x = 5$?

- A) -3
- B) 3
- C) 15
- D) 27

AF4-016

Question:

Solve: $9x - 7 = 4x + 18$

- A) 4
- B) 5
- C) 6
- D) 7

AF4-017

Question:

Which value of x makes $3x + 4$ greater than 19?

- A) 4
- B) 5
- C) 6
- D) 3

AF4-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : -2, 2, 6, 10

What is the rule?

- A) $y = 4x - 2$
- B) $y = 2x - 4$
- C) $y = -2x + 4$
- D) $y = 4x + 2$

AF4-019

Question:

Solve: $6(x + 2) = 3x + 30$

A) 4

B) 5

C) 6

D) 8

AF4-020

Question:

If $r = -2$ and $s = 7$, what is $4r + s$?

A) -15

B) -1

C) 1

D) 15

AF4-021

Question:

Which point is located in Quadrant IV?

A) (5, -3)

B) (-5, 3)

C) (-5, -3)

D) (5, 3)

AF4-022

Question:

A crew loads 32 crates in 4 hours. At the same rate, how many crates will the crew load in 9 hours?

A) 64

B) 72

C) 80

D) 96

AF4-023

Question:

Solve for m : $R = 7m - 14$

A) $m = R + 14$

B) $m = (R + 14)/7$

C) $m = 7(R + 14)$

D) $m = R/7 - 14$

AF4-024

Question:

What is the slope of a line that goes through (1, 5) and (5, 17)?

A) 2

B) 3

C) 4

D) 12

AF4-025

Question:

Simplify: $8x - 9 + 5x + 2$

- A) $3x - 7$
- B) $13x - 7$
- C) $13x + 11$
- D) $3x + 11$

AF4-026

Question:

If $g(x) = x^2 + 4x$, what is $g(4)$?

- A) 16
- B) 24
- C) 32
- D) 36

AF4-027

Question:

A sequence follows the rule “multiply by 3, then add 2.” If the first term is 4, what is the second term?

- A) 10
- B) 12
- C) 14
- D) 16

AF4-028

Question:

Solve: $4(x - 1) + 6 = 26$

- A) 4
- B) 5
- C) 6
- D) 7

AF4-029

Question:

A line crosses the y-axis at -6 and rises 2 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -6x + 2$
- B) $y = 2x - 6$
- C) $y = -2x - 6$
- D) $y = 6x + 2$

AF4-030

Question:

If $6/9 = x/27$, what is x ?

- A) 12
- B) 15
- C) 18
- D) 21

AF4-031

Question:

Solve: $-6x + 9 = -27$

- A) -6

- B) -3
- C) 3
- D) 6

AF4-032

Question:

A bin starts with 250 small parts. Each kit uses 12 parts. Which expression gives the number of parts left after k kits are made?

- A) $250 + 12k$
- B) $250 - 12k$
- C) $12k - 250$
- D) $250 \div 12k$

AF4-033

Question:

Which equation has $x = 7$ as its solution?

- A) $2x + 8 = 24$
- B) $5x - 4 = 31$
- C) $x/7 + 5 = 7$
- D) $3x + 6 = 30$

PART B — ANSWER KEY + EXPLANATIONS

AF4-001 — Correct: B — Explanation:

Solve $6x + 5 = 41$. Subtract 5 from both sides: $6x = 36$. Divide by 6: $x = 6$. The most common mistake is subtracting 5 correctly but forgetting to divide by 6.

AF4-002 — Correct: C — Explanation:

Start with $4n - 9 = 19$. Add 9 to both sides: $4n = 28$. Divide by 4: $n = 7$. A common mistake is subtracting 9 instead of adding it.

AF4-003 — Correct: A — Explanation:

Distribute first: $5(3x - 2) = 15x - 10$. Then subtract $4x$: $15x - 4x - 10 = 11x - 10$. A common mistake is forgetting to multiply -2 by 5.

AF4-004 — Correct: B — Explanation:

The item cost is \$14 per item, so that part is $14k$. The \$10 service fee is added once. The total cost is $14k + 10$. A common mistake is treating the service fee as if it repeats for each item.

AF4-005 — Correct: B — Explanation:

Substitute 3 for x : $f(3) = 6(3) - 5 = 18 - 5 = 13$. A common mistake is adding 5 instead of subtracting it.

AF4-006 — Correct: C — Explanation:

Test the choices in $y = 5x + 2$. For $(3, 17)$, $5(3) + 2 = 15 + 2 = 17$, which matches the y -value. A common mistake is choosing a point that is close but not exact.

AF4-007 — Correct: D — Explanation:

Solve $5(x - 3) = 30$. Divide both sides by 5: $x - 3 = 6$. Add 3: $x = 9$. A common mistake is adding 3 before undoing the multiplication by 5.

AF4-008 — Correct: C — Explanation:

The pattern increases by 5 each time: 2, 7, 12, 17. Add 5 to 17: 22. A common mistake is assuming the next number is 21 because the numbers are near multiples of 7.

AF4-009 — Correct: D — Explanation:

$x/8 = 7$ means x divided by 8 equals 7. Multiply both sides by 8: $x = 56$. A common mistake is adding 8 and 7 instead of multiplying.

AF4-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -3 and the y-intercept is 4, so the equation is $y = -3x + 4$. A common mistake is switching the slope and y-intercept.

AF4-011 — Correct: C — Explanation:

Find the unit rate: $84 \text{ pieces} \div 7 \text{ containers} = 12 \text{ pieces per container}$. For 11 containers: $12 \times 11 = 132$. A common mistake is adding 4 containers and 4 pieces instead of using the rate.

AF4-012 — Correct: B — Explanation:

Solve $20 - 4x = 8$. Subtract 20 from both sides: $-4x = -12$. Divide by -4: $x = 3$. A common mistake is losing the negative sign after subtracting 20.

AF4-013 — Correct: A — Explanation:

Combine like terms: $14y - 6y = 8y$. The constant 3 stays the same. The expression becomes $8y + 3$. A common mistake is subtracting the constant from the variable terms.

AF4-014 — Correct: B — Explanation:

The worker starts with 18 completed tasks. Then the worker completes 4 more each hour. After h hours, the total is $T = 4h + 18$. A common mistake is making 18 the hourly rate.

AF4-015 — Correct: A — Explanation:

Substitute $x = 5$ into $y = -3x + 12$: $y = -3(5) + 12 = -15 + 12 = -3$. A common mistake is treating $-3(5)$ as positive 15.

AF4-016 — Correct: B — Explanation:

Solve $9x - 7 = 4x + 18$. Subtract $4x$: $5x - 7 = 18$. Add 7: $5x = 25$. Divide by 5: $x = 5$. A common mistake is adding 7 to only one side.

AF4-017 — Correct: C — Explanation:

Solve $3x + 4 > 19$. Subtract 4: $3x > 15$. Divide by 3: $x > 5$. The only listed value greater than 5 is 6. A common mistake is choosing 5, but $3(5) + 4 = 19$, not greater than 19.

AF4-018 — Correct: A — Explanation:

The y-values increase by 4 each time x increases by 1, so the slope is 4. When $x = 0$, $y = -2$, so the y-intercept is -2. The rule is $y = 4x - 2$. A common mistake is using +2 instead of -2.

AF4-019 — Correct: C — Explanation:

Distribute: $6x + 12 = 3x + 30$. Subtract $3x$: $3x + 12 = 30$. Subtract 12: $3x = 18$. Divide by 3: $x = 6$. A common mistake is forgetting to multiply 2 by 6.

AF4-020 — Correct: B — Explanation:

Substitute $r = -2$ and $s = 7$: $4r + s = 4(-2) + 7 = -8 + 7 = -1$. A common mistake is treating $4(-2)$ as positive 8.

AF4-021 — Correct: A — Explanation:

Quadrant IV has a positive x-value and a negative y-value. The point (5, -3) fits that pattern. A common mistake is choosing (-5, 3), which is Quadrant II.

AF4-022 — Correct: B — Explanation:

32 crates in 4 hours means $32 \div 4 = 8$ crates per hour. In 9 hours: $8 \times 9 = 72$. A common mistake is multiplying 32 by 9 without finding the hourly rate first.

AF4-023 — Correct: B — Explanation:

Start with $R = 7m - 14$. Add 14 to both sides: $R + 14 = 7m$. Divide by 7: $m = (R + 14)/7$. A common mistake is dividing R by 7 first and then adding 14.

AF4-024 — Correct: B — Explanation:

Slope equals change in y divided by change in x. From (1, 5) to (5, 17), y increases by 12 and x increases by 4. The slope is $12 \div 4 = 3$. A common mistake is using 12 as the slope without dividing by the change in x.

AF4-025 — Correct: B — Explanation:

Combine like terms: $8x + 5x = 13x$, and $-9 + 2 = -7$. The simplified expression is $13x - 7$. A common mistake is writing $13x + 7$ because the negative sign before 9 is missed.

AF4-026 — Correct: C — Explanation:

Substitute 4 for x: $g(4) = 4^2 + 4(4) = 16 + 16 = 32$. A common mistake is calculating 4^2 as 8 instead of 16.

AF4-027 — Correct: C — Explanation:

Start with 4. Multiply by 3: 12. Then add 2: 14. A common mistake is adding 2 first and then multiplying.

AF4-028 — Correct: C — Explanation:

Solve $4(x - 1) + 6 = 26$. Distribute: $4x - 4 + 6 = 26$. Combine: $4x + 2 = 26$. Subtract 2: $4x = 24$. Divide by 4: $x = 6$. A common mistake is forgetting to combine -4 and +6.

AF4-029 — Correct: B — Explanation:

The line rises 2 units for every 1 unit to the right, so the slope is 2. It crosses the y-axis at -6, so the equation is $y = 2x - 6$. A common mistake is switching the slope and intercept.

AF4-030 — Correct: C — Explanation:

Use the proportion $6/9 = x/27$. Since $9 \times 3 = 27$, multiply 6 by 3: $x = 18$. A common mistake is adding 18 to the numerator because 9 becomes 27.

AF4-031 — Correct: D — Explanation:

Solve $-6x + 9 = -27$. Subtract 9 from both sides: $-6x = -36$. Divide by -6: $x = 6$. A common mistake is giving -6 because both sides contain negative numbers.

AF4-032 — Correct: B — Explanation:

The bin starts with 250 parts. Each kit uses 12 parts, so k kits use 12k parts. The number left is $250 - 12k$. A common mistake is adding 12k even though parts are being removed.

AF4-033 — Correct: B — Explanation:

Test $x = 7$. In choice B, $5(7) - 4 = 35 - 4 = 31$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.